CYS 410: SECURITY AND RISK MANAGEMENT PRACTICES

MS. NANYANGWE L.R

nanyangweleah@gmail.com

0967203891 (WHATSAPP)

COURSE CONTENT

- Introduction to Security Management
- Introduction to Psychology And Crime
- Criminology and Criminal Justice Systems
- Crime Prevention
- Legal Issues in Security Management
- Principles of Management
- Physical Security Management
- Institutional Security Management
- Introduction to Risk Management
- Threat Assessment Process
- Vulnerability Issues
- Risk Process
- Tools and Types of Risk Assessment
- The future of the risk management

READING LIST

1. Douglas E. Comer, Computer Networks and Internets with Internet Applications, Pearson Education Inc, 2007
2. Behrouz A. Forouzan, Data Communications and Netwoking, 4th Edition, McGrawHill, 2007
3. Andreas Von Grebmer, Information and IT Risk Management in a Nutshell: A pragmatic approach to Information Security, 2008, Books On Demand Gmbh

- Assignments (%)
- Tests/Quizzes (%)
- Final Exam (%)

INTRODUCTION TO SECURITY MANAGEMENT

- Security concepts and frameworks
- Security management lifecycle
- Role of security management in IT and business

WHAT IS SECURITY MANAGEMENT?

- Is the practice of protecting an organization's information systems and assets from threats, vulnerabilities, and unauthorized access.

- It involves assessing the organization's vulnerabilities, establishing appropriate security measures, and continuously monitoring and improving the security posture.
- It involves policies, procedures, and technologies designed to ensure confidentiality, integrity, and availability (CIA) of information.

GOAL OF SECURITY MANAGEMENT

- To protect an organization from internal and external threats, including theft, fraud, cyber-attacks, natural disasters, and terrorism.

KEY SECURITY CONCEPTS:

1. **Confidentiality** – Ensuring information is accessible only to those with proper authorization.
2. **Integrity** – Maintaining accuracy and reliability of data and preventing unauthorized alterations.
3. **Availability** – Ensuring information and resources are accessible when needed.
4. **Authentication** – Verifying the identity of users before granting access.
5. **Authorization** – Granting appropriate access rights based on user roles.
6. **Non-repudiation** – Ensuring that actions performed on a system can be traced back to a specific individual or entity.

COMMON SECURITY FRAMEWORKS:

- Security frameworks provide structured approaches to managing security risks. Some widely used frameworks include:

1. **ISO/IEC 27001** – International standard for information security management systems (ISMS).
2. **NIST Cybersecurity Framework (CSF)** – U.S. National Institute of Standards and Technology (NIST) guidelines for risk-based security.
3. **COBIT (Control Objectives for Information and Related Technologies)** – Governance framework for IT management and security.
4. **CIS Controls (Center for Internet Security)** – Set of best practices for cyber defence.

TYPES OF SECURITY

- Each security type focuses on different aspects of an organization's security, including:
  - Physical security
  - Information security
  - Cybersecurity
  - Personnel security
  - Operational security
  - Financial security

PHYSICAL SECURITY

- Physical security refers to the measures taken to protect an organization's people, assets, and facilities from physical threats such as theft, vandalism, and terrorism. Physical security measures may include access control systems, surveillance cameras, security guards, and perimeter fencing.

INFORMATION SECURITY

- Information security is concerned with protecting an organization's digital assets, including data, software, and hardware, from unauthorized access, use, disclosure, modification, or destruction. Information security measures may include firewalls, antivirus software, encryption, and access controls.

OBJECTIVES OF INFORMATION SECURITY

- The three objectives of information security are:
  - Confidentiality
  - Integrity
  - Availability

CONFIDENTIALITY

- Confidentiality is the practice of keeping information safe and private.
- Information must be protected from unauthorized access by third parties. It is important to highlight exactly who is authorised to prevent any valuable information from being accessed.

MEASURES THAT SEEK TO PROTECT THE CONFIDENTIALITY OF INFORMATION INCLUDE:

- Encryption of data
- User access control
- Physical and environmental security
- Operational security
- Communications security

INTEGRITY

- Integrity means that data/information is protected from being changed (either unintentionally or by unauthorised parties), thus being ‘reliable’.
- Integrity primarily means protection against unintentional changes that may likely occur due to defective systems.

MEASURES THAT ASCERTAIN INTEGRITY INCLUDE:

- Asset Management
- User control
- System acquisition, development, and maintenance

AVAILABILITY

- Availability means building the technological infrastructure that makes data and information available.

MEASURES THAT SEEK TO PROTECT THE AVAILABILITY OF INFORMATION INCLUDE:

- Risk analysis
- System acquisition, development, and maintenance
- Incident management
- Business continuity management

CYBERSECURITY

- Cybersecurity is a subset of information security that specifically deals with protecting an organization's digital assets from cyber threats, including hackers, viruses, and other malicious software.

PERSONNEL SECURITY

- Personnel security involves measures to ensure that an organization's employees and contractors are trustworthy and reliable. Personnel security measures may include background checks, security clearances, and ongoing monitoring of employee behavior.

OPERATIONAL SECURITY

- Operational security is concerned with protecting an organization's operations from disruption or compromise. Operational security measures may include business continuity planning, crisis management, and emergency response planning.

FINANCIAL SECURITY

- Financial security refers to measures taken to protect an organization's financial assets from fraud, theft, or other forms of financial crimes. Financial security measures may include auditing, fraud detection, and risk management. personnel security, operational security, and financial security

SECURITY MANAGEMENT LIFECYCLE

- The security management lifecycle ensures continuous protection of an organization's assets through a structured process. It typically consists of the following phases:

SECURITY MANAGEMENT LIFECYCLE…

1. **Identification** – Recognizing assets, vulnerabilities, and threats.
2. **Assessment** – Evaluating risks and potential impacts.
3. **Planning & Policy Development** – Establishing security policies, standards, and guidelines.
4. **Implementation** – Deploying security controls, such as firewalls, encryption, and access controls.

SECURITY MANAGEMENT LIFECYCLE…

5. **Monitoring & Detection** – Continuously tracking system activity to identify anomalies or threats.
6. **Incident Response** – Reacting to security breaches with containment, eradication, and recovery measures.
7. **Review & Improvement** – Assessing security measures and updating policies based on emerging threats.

ROLE OF SECURITY MANAGEMENT IN IT

- **Protecting Data and Systems** – Prevents unauthorized access and cyber threats.
- **Ensuring System Availability** – Maintains business continuity through disaster recovery planning.
- **Regulatory Compliance** – Adheres to data protection laws such as GDPR, HIPAA, and PCI-DSS.
- **Incident Management** – Identifies and responds to security breaches effectively.

ROLE OF SECURITY MANAGEMENT IN IT

- **Risk Management** – Identifies and mitigates risks that could disrupt business operations.
- **Brand Reputation** – A strong security posture builds customer trust and credibility.
- **Operational Continuity** – Prevents financial and operational losses from cyber incidents.
- **Competitive Advantage** – Companies with robust security policies attract more customers and partners.

PRACTICAL EXERCISE: DEVELOPING AN INFORMATION SECURITY POLICY

Scenario:

- You have been hired as a security consultant to develop an Information Security Policy for an NGO. Your task is to create a policy document outlining key security measures based on the ISO 27001 framework.

Instructions:

- Clearly define acceptable use, access control, incident response and Compliance & Enforcement procedures.
- Align the policy with the ISO 27001 standard to ensure best practices.
- Use simple, professional language suitable for employees at all levels.

INTRODUCTION TO PSYCHOLOGY AND CRIME

- Security and crime prevention are not just about technology; human behaviour plays a crucial role in understanding criminal activities, especially in cybersecurity.
- Psychology helps explain why individuals commit crimes and how security professionals can anticipate and mitigate threats.
  - Theories of criminal behaviour
  - Psychological factors in cybercrime
  - Social engineering and cybersecurity threats

THEORIES OF CRIMINAL BEHAVIOUR

- Criminal behaviour theories help explain why people engage in illegal activities, including cybercrime. Some key theories include:
  A. Classical Criminology Theory
  B. Psychological Theories
  C. Sociological Theories
  D. Routine Activity Theory

A. CLASSICAL CRIMINOLOGY THEORY

- Proposed by Cesare Beccaria and Jeremy Bentham.
- Assumes people commit crimes based on rational decision-making.
- Cybercriminals, such as hackers, may weigh the risks and rewards before engaging in illegal activities.

B. PSYCHOLOGICAL THEORIES

- **Personality Theory** – Suggests that personality traits (e.g., impulsivity, aggression, antisocial behaviour) influence criminal behaviour.
- **Psychoanalytic Theory (Freud)** – Suggests unresolved psychological conflicts or childhood trauma may lead individuals to commit crimes.

C. SOCIOLOGICAL THEORIES

- **Strain Theory (Robert Merton)** – People turn to crime when they experience frustration from failing to achieve societal success.
  - Example: Hackers engaging in financial fraud due to economic hardship.
- **Social Learning Theory (Albert Bandura)** – Criminal behaviour is learned from observing others.
  - Example: A novice cybercriminal learning hacking techniques from online forums.

D. ROUTINE ACTIVITY THEORY

- Crime occurs when three factors align:
  1. Motivated offender
  2. Suitable target
  3. Lack of capable guardian (security controls)
- Example: Phishing attacks succeed when users lack awareness and proper security defences.

2. PSYCHOLOGICAL FACTORS IN CYBERCRIME

- Psychological motivations influence why people engage in cybercrime. Common psychological factors include:
  A. Greed and Financial Gain
  B. Revenge and Retaliation
  C. Thrill-Seeking and Curiosity
  D. Ideology and Hacktivism
  E. Mental Disorders and Impulsivity
  F. Social Engineering and Cybersecurity Threats

A. GREED AND FINANCIAL GAIN

- Many cybercriminals engage in hacking, fraud, and ransomware attacks for financial profit.
  - Example: Cybercriminals use phishing scams to steal banking details.

B. REVENGE AND RETALIATION

- Employees or insiders may conduct cyberattacks to get back at an organization.
  - Example: A fired employee launching a denial-of-service (DoS) attack against a company's website.

C. THRILL-SEEKING AND CURIOSITY

- Some hackers are motivated by the challenge and excitement of breaking into secure systems.
  - Example: "Script kiddies" testing hacking tools to deface websites.

D. IDEOLOGY AND HACKTIVISM

- Hackers may commit cybercrimes for political, social, or religious reasons.
  - Example: Hacktivist groups like Anonymous launching cyberattacks against government institutions.

E. MENTAL DISORDERS AND IMPULSIVITY

- Some criminals may have psychological disorders (e.g., antisocial personality disorder) that drive them to commit cybercrimes without remorse.

3. SOCIAL ENGINEERING AND CYBERSECURITY THREATS

- Social engineering is a psychological manipulation technique used by cybercriminals to deceive people into revealing sensitive information or performing actions that compromise security.

COMMON SOCIAL ENGINEERING ATTACKS:

1. **Phishing** – Fraudulent emails pretending to be from legitimate sources to steal login credentials.
2. **Spear Phishing** – Targeted phishing attacks against specific individuals or organizations.
3. **Pretexting** – Creating a fake scenario to trick people into giving out sensitive data.
4. **Baiting** – Offering something enticing (e.g., free software) that installs malware.
5. **Quid Pro Quo** – Attackers promise rewards (e.g., free IT support) in exchange for access.
6. **Tailgating/Piggybacking** – Gaining unauthorized access by following an authorized person into a secure area.

PSYCHOLOGICAL TACTICS USED IN SOCIAL ENGINEERING:

- **Authority** – Pretending to be a boss or IT support staff.
- **Urgency** – Creating pressure to act quickly.
- **Fear and Intimidation** – Threatening legal action or account closure.
- **Reciprocity** – Offering a favour in exchange for information.

HOW TO MITIGATE PSYCHOLOGICAL CYBER THREATS?

1. **User Awareness Training** – Educating employees and users about social engineering tactics.
2. **Multi-Factor Authentication (MFA)** – Preventing unauthorized access even if credentials are stolen.
3. **Email Filtering and Anti-Phishing Tools** – Blocking malicious emails before they reach users.
4. **Strict Security Policies** – Implementing strong authentication and verification processes.
5. **Behavioural Monitoring** – Detecting unusual user activities that indicate security threats.

3. CRIMINOLOGY AND CRIMINAL JUSTICE SYSTEMS

- Criminology is the study of crime, criminal behaviour, and the societal response to crime. The criminal justice system encompasses the institutions and processes established to prevent, investigate, and punish criminal activities, including cybercrime.
  - Overview of criminology
  - Criminal justice processes
  - Cybercrime laws and enforcement

1. OVERVIEW OF CRIMINOLOGY

- Criminology examines the causes, consequences, and prevention of crime. It integrates sociology, psychology, law, and forensic science to understand criminal behaviour.

KEY AREAS OF CRIMINOLOGY

1. **Theories of Crime** – Explains why individuals commit crimes (e.g., strain theory, social learning theory).
2. **Types of Crime** – Includes violent crime, white-collar crime, organized crime, and cybercrime.
3. **Crime Prevention Strategies** – Involves law enforcement, social policies, and technological measures to reduce crime.

BRANCHES OF CRIMINOLOGY

- **Penology** – Studies punishment, prisons, and rehabilitation.
- **Victimology** – Examines crime victims and their interactions with criminals and law enforcement.
- **Forensic Criminology** – Uses scientific methods to analyze crimes.
- **Cyber Criminology** – Focuses on digital crimes like hacking, identity theft, and cyberterrorism.

2. CRIMINAL JUSTICE PROCESSES

- The criminal justice system consists of institutions and procedures designed to enforce laws, prosecute criminals, and ensure justice.

KEY COMPONENTS OF THE CRIMINAL JUSTICE SYSTEM

1. **Law Enforcement (Police and Investigators)**

- First responders to crimes.
- Conduct investigations and gather evidence.
- Enforce laws and maintain public safety.

KEY COMPONENTS OF THE CRIMINAL JUSTICE SYSTEM…

- Judicial System (Courts and Prosecutors)
- Courts determine guilt or innocence.
- Prosecutors and defence attorneys present cases.
- Judges and juries decide verdicts and sentencing.

KEY COMPONENTS OF THE CRIMINAL JUSTICE SYSTEM…

- Correctional System (Prisons and Rehabilitation)
- Manages convicted criminals.
- Includes imprisonment, probation, and rehabilitation programs.
- Aims to reform offenders and prevent repeat crimes.

3. CYBERCRIME LAWS AND ENFORCEMENT

- Cybercrime refers to illegal activities conducted through digital means, including hacking, fraud, and cyberterrorism.

TYPES OF CYBERCRIME

- Computer-Related Crimes – Hacking, malware attacks, denial-of-service (DoS) attacks.
- Financial Cybercrimes – Online fraud, identity theft, phishing.
- Cyber Harassment and Exploitation – Cyberbullying, revenge porn, child exploitation.
- Cyberterrorism and Espionage – Attacks on critical infrastructure, data breaches.

KEY CYBERCRIME LAWS AND REGULATIONS

- Laws vary by country, but some common international frameworks include:

1. **The Budapest Convention on Cybercrime (2001)**
   - First international treaty addressing cybercrime.
   - Covers hacking, fraud, child exploitation, and copyright violations.
   - Encourages cross-border cooperation in cybercrime investigations.
2. **The Computer Fraud and Abuse Act (CFAA, USA)**
   - Criminalizes unauthorized access to computer systems.

KEY CYBERCRIME LAWS AND REGULATIONS…

3. **The General Data Protection Regulation (GDPR, EU)**
   - Protects user data and privacy rights.
   - Penalizes organizations for mishandling personal data.
4. **Zambia's Cyber Security and Cyber Crimes Act (2021)**
   - Addresses hacking, online fraud, identity theft, and electronic evidence handling.
   - Provides for digital forensics and law enforcement collaboration.

CYBERCRIME ENFORCEMENT AGENCIES

- **Interpol Cybercrime Unit** – Coordinates international efforts to combat cybercrime.
- **Federal Bureau of Investigation (FBI – Cyber Division, USA)** – Investigates cyber threats, financial fraud, and terrorism-related cyber activities.
- **Europol's European Cybercrime Centre (EC3)** – Supports cybercrime investigations in EU countries.
- **Zambia Information and Communications Technology Authority (ZICTA)** – Enforces cyber laws and protects digital infrastructure.

CHALLENGES IN CYBERCRIME ENFORCEMENT

- **Anonymity of Cybercriminals** – Attackers use VPNs, encryption, and the dark web.
- **Jurisdiction Issues** – Cybercrimes often cross-national orders.
- **Rapidly Evolving Technology** – Law enforcement struggles to keep up.
- **Lack of Awareness and Reporting** – Many cybercrimes go unreported due to stigma or lack of understanding.

CYBERCRIME PREVENTION STRATEGIES

- Strong cybersecurity laws and enforcement.
- Public awareness campaigns on cyber threats.
- Collaboration between governments, private sectors, and tech companies.
- Implementation of cybersecurity frameworks like NIST, ISO 27001.

4. CRIME PREVENTION

- Crime prevention refers to strategies, policies, and techniques aimed at reducing criminal activities and minimizing their impact on individuals and society. In the digital age, crime prevention also includes cybersecurity awareness and fraud prevention techniques.
- Crime prevention models
- Cybersecurity awareness strategies
- Fraud prevention techniques

1. CRIME PREVENTION MODELS

- Crime prevention models provide structured approaches to reducing crime through various interventions and strategies.

I. CRIME PREVENTION MODELS…

- **A. Situational Crime Prevention (SCP)**
  - Focuses on **reducing opportunities** for crime by altering the environment.
  - Uses techniques such as **surveillance, access control, and security enhancements** to deter criminals.
  - Example: CCTV cameras, better street lighting, alarm systems.

1. CRIME PREVENTION MODELS…

- **B. Crime Prevention Through Environmental Design (CPTED)**
- Uses **urban design and physical environments** to reduce crime.
- Key principles:
  1. **Natural Surveillance** – Increasing visibility to deter criminals (e.g., open spaces, clear sightlines).
  2. **Access Control** – Restricting entry to unauthorized individuals (e.g., gates, keycard access).
  3. **Territorial Reinforcement** – Clearly marking spaces to show ownership and responsibility (e.g., fences, signs).

I. CRIME PREVENTION MODELS…

- **C. Social Crime Prevention**
- Focuses on addressing **root causes of crime**, such as poverty, lack of education, and unemployment.
- Strategies include **community programs, youth engagement, and education initiatives** to prevent crime before it happens.

I. CRIME PREVENTION MODELS…

- **D. Law Enforcement-Based Crime Prevention**
- Involves **policing strategies and legal frameworks** to deter and punish crime.
- Methods:
  - **Community policing** – Officers work with communities to build trust and prevent crime.
  - **Zero-tolerance policies** – Strict enforcement of laws to deter minor crimes before they escalate.

2. CYBERSECURITY AWARENESS STRATEGIES

- Cybersecurity awareness is crucial in preventing cybercrimes such as hacking, identity theft, and online fraud.

2. CYBERSECURITY AWARENESS STRATEGIES…

- **A. Education and Training**
  - Conducting **cybersecurity awareness workshops** for employees and the public.
  - Implementing **mandatory security training** in organizations.
  - Using **real-life cyberattack simulations** to educate people about potential threats.

2. CYBERSECURITY AWARENESS STRATEGIES…

- **B. Safe Online Practices**
  - **Password Management** – Using strong passwords and multi-factor authentication (MFA).
  - **Phishing Awareness** – Recognizing fraudulent emails and links.
  - **Secure Browsing** – Avoiding suspicious websites and downloads.
  - **Data Protection** – Encrypting sensitive data and backing up files regularly.

2. CYBERSECURITY AWARENESS STRATEGIES…

- **C. Security Frameworks and Policies**
  - Organizations implement standards like ISO 27001, NIST Cybersecurity Framework, and GDPR to protect data and prevent cyber threats.
  - Governments enforce cybercrime laws to regulate online activities and enhance security.

3. FRAUD PREVENTION TECHNIQUES

- Fraud prevention is critical in minimizing financial crimes such as identity theft, credit card fraud, and corporate fraud.

3. FRAUD PREVENTION TECHNIQUES…

- **A. Identity Verification and Authentication**
  - Multi-Factor Authentication (MFA) – Requires users to verify identity through multiple steps (e.g., passwords, biometrics).
  - Know Your Customer (KYC) Policies – Used by banks and financial institutions to verify customer identities.

3. FRAUD PREVENTION TECHNIQUES…

- **B. Transaction Monitoring and Fraud Detection**
  - AI and Machine Learning Algorithms – Detect unusual transaction patterns and flag suspicious activities.
  - Geolocation Tracking – Prevents unauthorized transactions from foreign locations.

3. FRAUD PREVENTION TECHNIQUES…

- **C. Employee and Customer Awareness**
  - Training employees to recognize fraudulent schemes like business email compromise (BEC) and fake invoicing.
  - Educating customers about online fraud risks and best practices.

3. FRAUD PREVENTION TECHNIQUES…

- **D. Legal and Regulatory Compliance**
  - Companies must comply with financial regulations like:
    - Anti-Money Laundering (AML) Laws
    - General Data Protection Regulation (GDPR)
    - The Payment Card Industry Data Security Standard (PCI DSS)

LEGAL ISSUES IN SECURITY MANAGEMENT

- Security management involves compliance with laws and regulations to ensure data protection, privacy, and ethical standards.
- Organizations must understand cybersecurity laws, data privacy regulations, and ethical responsibilities to mitigate legal risks.

- Cybersecurity laws and regulations
- Data privacy laws (e.g., GDPR, HIPAA)
- Ethical considerations in security

1. CYBERSECURITY LAWS AND REGULATIONS

- Cybersecurity laws are designed to protect information systems, critical infrastructure, and personal data from cyber threats.

A. KEY CYBERSECURITY LAWS

1. Computer Fraud and Abuse Act (CFAA) – USA
   - Prohibits unauthorized access to computer systems.
   - Criminalizes hacking, fraud, and cyber espionage.
2. General Data Protection Regulation (GDPR) – EU
   - Enforces strict rules on data protection and privacy.
   - Requires organizations to report data breaches within 72 hours.
   - Grants individuals the "right to be forgotten."

A. KEY CYBERSECURITY LAWS…

3. Cybersecurity Act (CSA) – EU
   - Establishes security standards for businesses and organizations.
   - Focuses on risk management and critical infrastructure protection.
4. Personal Data Protection Act (PDPA) – Singapore
   - Regulates data collection, processing, and protection.
   - Requires organizations to obtain consent before collecting personal information.
5. Zambia's Cyber Security and Cyber Crimes Act (2021)
   - Criminalizes cybercrime, including hacking, online fraud, and cyber harassment.
   - Establishes penalties for identity theft and unauthorized access to data.

B. INDUSTRY-SPECIFIC CYBERSECURITY LAWS

- The Health Insurance Portability and Accountability Act (HIPAA) – USA
  - Protects medical data and ensures patient privacy.
- The Payment Card Industry Data Security Standard (PCI DSS)
  - Regulates payment processing and protects credit card transactions.

2. DATA PRIVACY LAWS

- Data privacy laws regulate how personal data is collected, stored, and shared.

A. GENERAL DATA PROTECTION REGULATION (GDPR) – EUROPE

- Applies to organizations handling EU citizens' data, regardless of location.
- Requires clear consent for data collection.
- Imposes heavy fines for non-compliance (up to 4% of annual revenue).

B. HEALTH INSURANCE PORTABILITY AND ACCOUNTABILITY ACT (HIPAA) – USA

- Protects medical records and personal health information.
- Requires encryption and secure storage of health data.
- Enforces penalties for breaches of patient confidentiality.

C. CALIFORNIA CONSUMER PRIVACY ACT (CCPA) – USA

- Grants consumers the right to **access, delete, and opt out** of data collection.
- Businesses must disclose what personal data they collect and how it’s used.

D. ZAMBIA'S DATA PROTECTION ACT

- Regulates the processing of personal data by organizations.
- Establishes data subject rights, such as access, correction, and deletion.

3. ETHICAL CONSIDERATIONS IN SECURITY

- Security professionals must follow ethical principles to ensure transparency, fairness, and accountability.

A. ETHICAL ISSUES IN SECURITY MANAGEMENT

- **Surveillance vs. Privacy** – Balancing security needs with individuals' rights to privacy.
- **Employee Monitoring** – Ethical concerns over tracking emails, keystrokes, and internet usage.
- **Use of AI in Security** – AI-driven surveillance and facial recognition raise bias and discrimination concerns.

B. ETHICAL GUIDELINES FOR SECURITY PROFESSIONALS

1. **Respect Privacy** – Handle sensitive data responsibly and minimize data collection.
2. **Transparency** – Clearly communicate security policies and data handling practices.
3. **Accountability** – Ensure security professionals take responsibility for breaches or ethical violations.
4. **Compliance with Laws** – Follow international and local cybersecurity regulations.

C. PROFESSIONAL ETHICS IN CYBERSECURITY

- **(ISC)² Code of Ethics** – Guides cybersecurity professionals in maintaining integrity.
- **ACM Code of Ethics** – Promotes responsible computing practices.